

Comparison Overview

A. Our Dataset advantages

- ⇒ Our paper does work at the product level (Open food facts, LCA, FAO) + MRIO dataset => which is why it's for consumer behavior insights.
- ⇒ New dataset in 2022

Data Flows:

- Open food fact data: Kaggle auth & OFF download (full dataset)
- Language detection & NLLB-200 translation
- Product names → MRIO sector mapping (fuzzy + semantic; scaffold)
- LCA data (Clark et al. + FAOSTAT): align to CO₂/Land/Water, unit harmonization
- MRIO data: EXIOBASE (Z, Y, Q) load & reshape; compute A, ****Leontief****
 $L=(I-A)^{-1}$; hybridize satellite with micro data

B. Other papers

1. Methodology Comparison

Feature	Bruckner et al. (2019) – FABIO	Clark et al. (2022)
Approach	Multi-Regional Input-Output (MRIO) Model	Product-level Life Cycle Assessment (LCA) + Machine Learning
Model Type	Supply-use based MRIO model	Bottom-up LCA, using a predictive model
Aggregation	National-level commodity flows	Individual food product level
Scope	Agriculture, food, and biomass globally	57,000 food products from UK and Ireland retail markets
Goal	Understand global biomass flows and their environmental impacts	Provide per-product environmental impact scores to guide consumers

2. Data Sources

Data Type	FABIO	Clark et al.
Trade & Supply-Use Data	FAOSTAT (crops/livestock), UN Comtrade, IEA, WHO, EIA, etc.	Not directly used; relies on retail product data
Food/Product Information	FAO commodities (~130 items, 191 countries, 1986–2013)	Ingredients & packaging info from Open Food Facts and brands
Other	Harmonization with EXIOBASE for hybrid modeling	LCA coverage mapped to product metadata using NLP + classifiers

3. Modeling Techniques

Category	FABIO	Clark et al.
Type of LCA	Environmentally Extended Input-Output (EEIO)	Process-based LCA
Machine Learning	No	Random Forest model to estimate impact from ingredients
Granularity	Sector/commodity level	Product-level (based on label/ingredient list)
Hybridization	Yes, linked with EXIOBASE to improve environmental extension	No hybridization, but extensive product-LCA mapping

⇒ **FABIO (Bruckner et al., 2019) is a top-down, sector-level model and does not operate at the level of individual products.**

What does that mean?

- **FABIO tracks flows of biomass** (e.g., crops, livestock, forestry products) **between sectors and countries**, not specific brands, food items, or retail products.
- It uses data like:
 - **Crop production** by country
 - **Trade flows** between countries
 - **Sectoral use** (e.g., how much wheat goes to food processing, biofuels, animal feed, etc.)
- The resolution is about **130 commodity categories** across **191 countries**, over multiple years.